BUILD MANUAL

Space-Debris Spin-State Photometric Testbed

Step-by-step construction, wiring, commissioning and first-light procedure. Every step carries its own verification check.

DOCUMENT	SDC-002 — Build Manual, Revision A
COMPANION	SDC-001 Rev B — Design & Build Specification (bill of materials, rationale)
BUILD TIME	Approximately 30 working hours across 6 days, excluding print time
SKILL LEVEL	Basic hand tools, soldering, and a multimeter. No machining required.
VERIFICATION	73 numbered steps, each with its own check; 8 hard gates between phases
STATUS	Geometry verified numerically — see Section 1.3

IMPORTANT — THIS MANUAL SUPERSEDES THE ARRANGEMENT SHOWN IN SDC-001
During preparation of this manual the swept envelope of the target was computed numerically for the first time. The straddle arrangement drawn in SDC-001 Rev B (Figs. 4 and 5) **does not work**: the target strikes the support posts by 87 mm on each side at large tilt angles. The mechanism has been rearranged as a cantilever, in which both bearings sit on one side of the sweep plane. Section 1.3 shows the analysis and the corrected geometry. **Build to this manual, not to the SDC-001 elevations.** All other content of SDC-001 — the bill of materials, electrical design, optical design and software — remains valid.

BEFORE YOU START

How to use this manual

Work through the steps in order. Each is written as a single action, followed by a shaded **CHECK** line describing exactly what must be true before you move on. Tick the box.

Between phases there are **verification gates**. A gate is not a formality — each one exists because the fault it catches becomes far more expensive to find later. If a gate fails, fix it there. Do not carry a known fault forward on the assumption that it will not matter.

— **Bold codes in brackets** such as [A01] refer to the bill of materials in SDC-001 Rev B.

— All dimensions are millimetres.

— Coordinates use X along the shaft axis, Y across it, Z upward from the bench.

— **Threadlock** means a single drop of medium-strength blue threadlocker [I07]. Every grub screw in this build gets one. Vibration will loosen anything that does not.

Safety

Hazard	Precaution
Rotating machinery	The target sweeps a 350 mm envelope at speed. Keep hands and cables clear. Never reach into the envelope with power applied — open SW1 first.
Electrical	The 12 V rail can deliver 5 A into a short. Fit F1 and SW1 before connecting anything else. Never work on wiring with the supply plugged in.
Driver damage	Inserting or removing a motor connector with 12 V applied destroys the driver through back-EMF. Open SW1 every time, without exception.
Eye	The 10 W COB LED is uncomfortably bright at close range. Do not look into it directly; position it before powering it.
Hot surfaces	The COB heatsink and the driver heatsinks run warm. Allow them to cool before handling.
Tools	Deburr cut extrusion — the edges are sharp. Wear eye protection when cutting.

Numerical verification of the design

The geometry in this manual has been checked by calculation rather than by eye. Ten checks were run; the results are quoted at the relevant steps and summarised here.

Check	Result	Status
Straddle arrangement, swept envelope vs posts	Envelope reaches ±177 mm; post inner faces at ±90 mm	**FAIL — rearranged**
Cantilever arrangement, envelope vs posts	Sweep plane at X=180; nearest post face at X=270; axial gap 60 mm	PASS
Envelope vs bench	Lowest point 160 mm above bench; rail top at 20 mm	PASS (140 mm)
Cradle stack height chain	Arm inner space 52 mm vs motor body 48 mm; shaft protrusion 18 mm vs hub 18 mm	PASS (4 mm clear)
Belt selection	300 mm (150 T) gives 109.3 mm centres; 3:1 ratio	PASS
Motor B clearance	Body spans Z 70–112 and X 395–443; base ends at 460	PASS
Cantilever bearing loads	10.6 N near, 6.2 N far; 608ZZ rated ≈3400 N	PASS (320×)
Shaft bending	8.7 MPa against ≈250 MPa yield; tip deflection 0.06 mm	PASS (29×)
Step-rate budget	1730 + 561 = 2291 steps/s against ≈4000 ceiling	PASS (57 %)
Enclosure envelope	Needs 500 × 401 × 400; specify 550 × 500 × 450	PASS

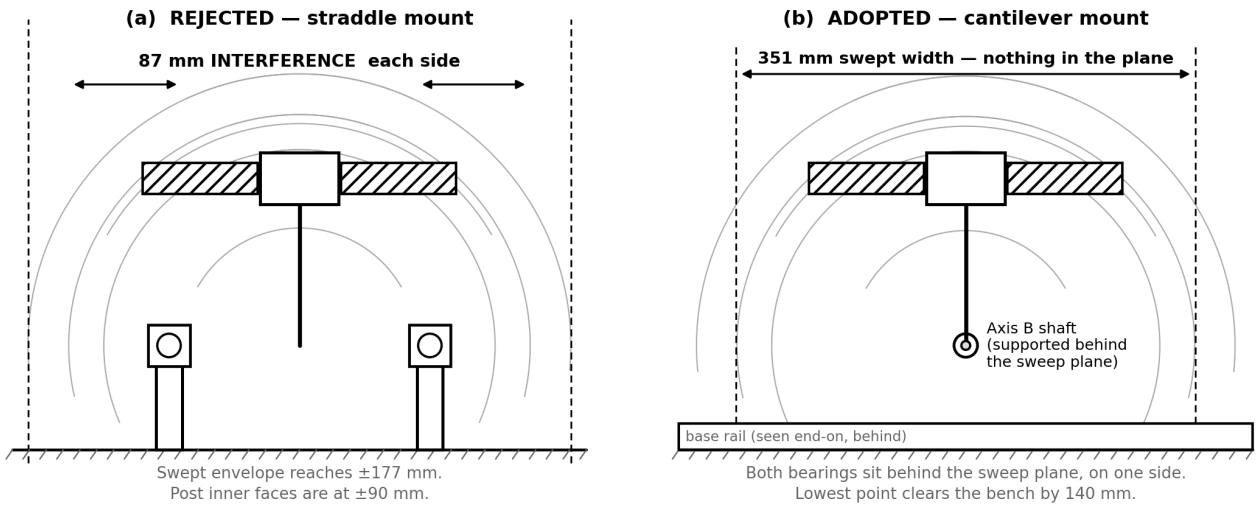


FIG. M1 — Why the arrangement changed. (a) The straddle mount drawn in SDC-001: the swept envelope of the target overlaps the support posts by 87 mm on each side. (b) The cantilever mount adopted here: both bearings sit behind the sweep plane, so nothing obstructs the target through the full ±60° range.

PHASE 0

Preparation and kitting

Do all of this before picking up a tool. Two of these steps run unattended for hours and belong at the very start of day 1.

1. Start both 3D prints — the two target buses, four panels, the cradle, and two bearing housings. Print PETG for the cradle and housings, PLA for the targets. [G01, C11]

CHECK [ ] Both prints running. First layer adhered cleanly across the whole footprint.

2. Lay out every part from the bill of materials and check it against the list. Group by phase into separate trays or bags.

CHECK [ ] Every line item accounted for. Any shortfall identified now, while there is still time to buy it.

3. Confirm you have all tools listed in SDC-001 Appendix B.4, in particular a working multimeter.

WARNING — Do not start the electrical phases without a multimeter. Coil identification and current setting are both impossible without one, and guessing at either will destroy components.

CHECK [ ] Multimeter reads a known resistor correctly on the ohms range.

4. Cut the extrusion to the cut list: two at 460 mm, two at 160 mm, two at 200 mm. Have this done at the shop if possible — a shop cut is squarer than a hand-sawn one. [C01–C03]

CHECK [ ] All six lengths within ±1 mm and square to within 0.5 mm across the face.

5. Deburr every cut end with a file or 400-grit paper.

WARNING — A burr prevents a bracket from seating square. The frame will then never come true, and no amount of later adjustment will fix it.

CHECK [ ] No burr catches a fingernail dragged across any cut face.

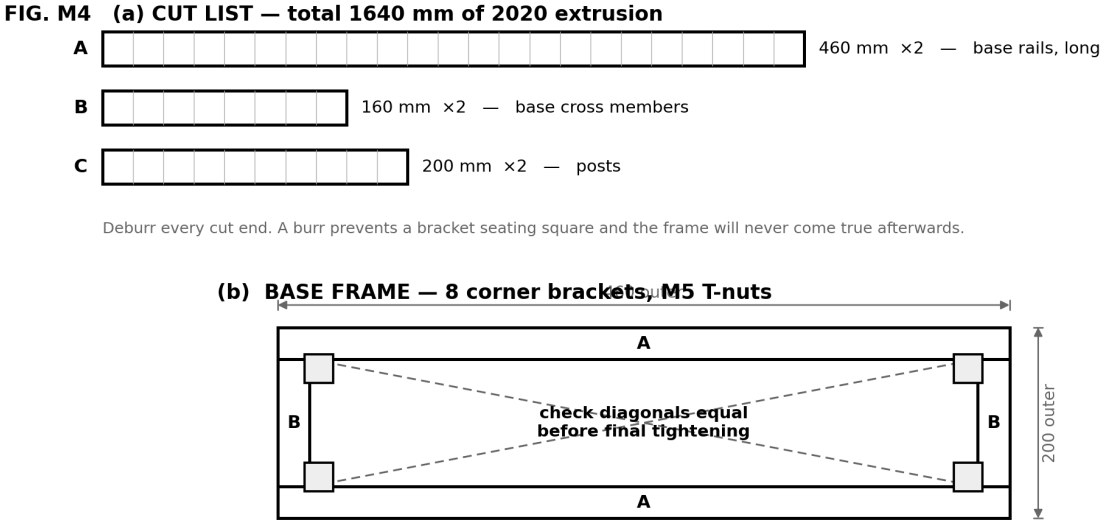


FIG. M4 — (a) Cut list, 1640 mm of 2020 extrusion in total. (b) Base frame layout with the diagonal check that must be made before final tightening.

PRE-BUILD CHECK — KITTING

All parts present and checked against the BOM. Six extrusion lengths cut, deburred and verified square. Prints running. Multimeter confirmed working.

If this gate fails, stop. A missing part discovered in Phase 3 costs a market trip and most of a working day. Finding it now costs nothing.

PHASE 1

Base frame and posts

FIG. M2 GENERAL ARRANGEMENT — PLAN VIEW (dimensions in mm)

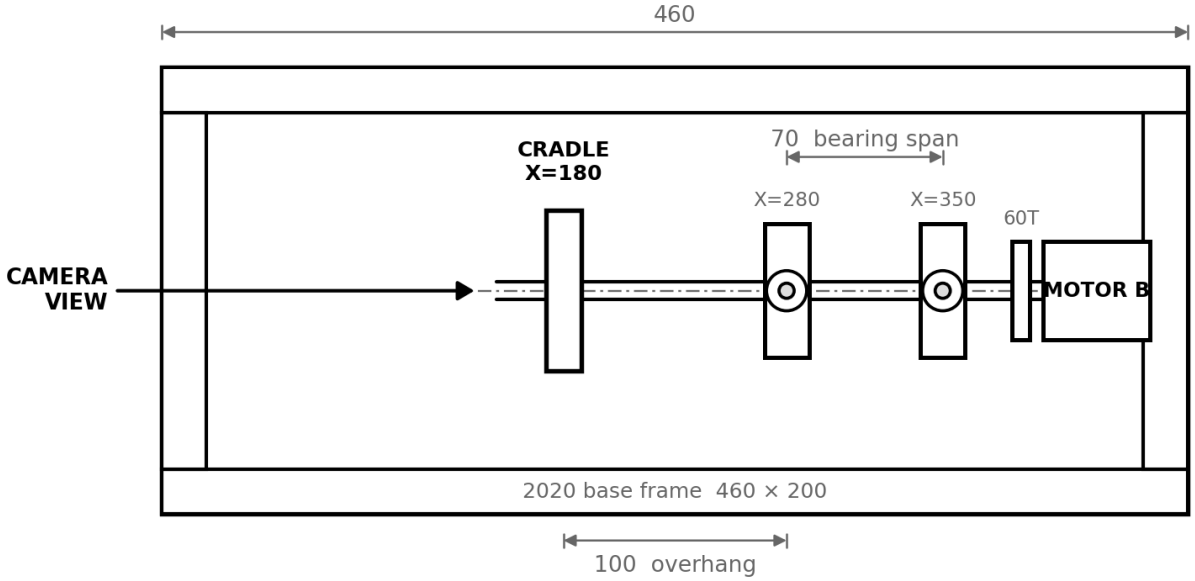


FIG. M2 — General arrangement in plan. The shaft runs along X. Both bearings are at X = 280 and X = 350; the cradle hangs at X = 180, giving a 100 mm overhang and 70 mm bearing span.

6. Assemble the base rectangle: two 460 mm rails and two 160 mm cross members, joined with eight corner brackets on M5 T-nuts. Leave all bolts finger-tight. [C01, C02, C04, C05, C06]

CHECK [ ] Outer dimensions measure 460 × 200 mm.

7. Measure both diagonals of the base. Adjust until they are equal, then tighten all eight brackets progressively, working around the frame rather than one corner at a time.

CHECK [ ] Diagonals equal to within 1 mm, and remain so after tightening.

8. Fit the four rubber feet to the underside, one near each corner. [C14]

CHECK [ ] The frame sits on the bench without rocking. Press each corner in turn — none moves.

9. Mount the two 200 mm posts vertically at X = 280 and X = 350, using two corner brackets each, bolted into the base rail's top slot. [C03, C04, C05, C06]

CHECK [ ] Both posts plumb in two planes, checked with a spirit level. Post tops at Z = 220 mm.

10. Check the post spacing at the top as well as at the base.

CHECK [ ] Post centres 70 ± 0.5 mm apart at both top and bottom — the posts are parallel, not splayed.

VERIFICATION GATE G1 — FRAME

Base square to within 1 mm on the diagonals. Frame does not rock. Both posts plumb and parallel, tops at Z = 220 mm. Pushing the top of either post by hand produces no visible deflection.

If this gate fails, stop. Everything downstream references these posts. A frame that is out of true puts the two bearings out of coaxial alignment, and the shaft will bind no matter how carefully the rest is built.

PHASE 2

Axis B — bearings, shaft and belt drive

FIG. M5 AXIS B BEARING AND SHAFT ASSEMBLY

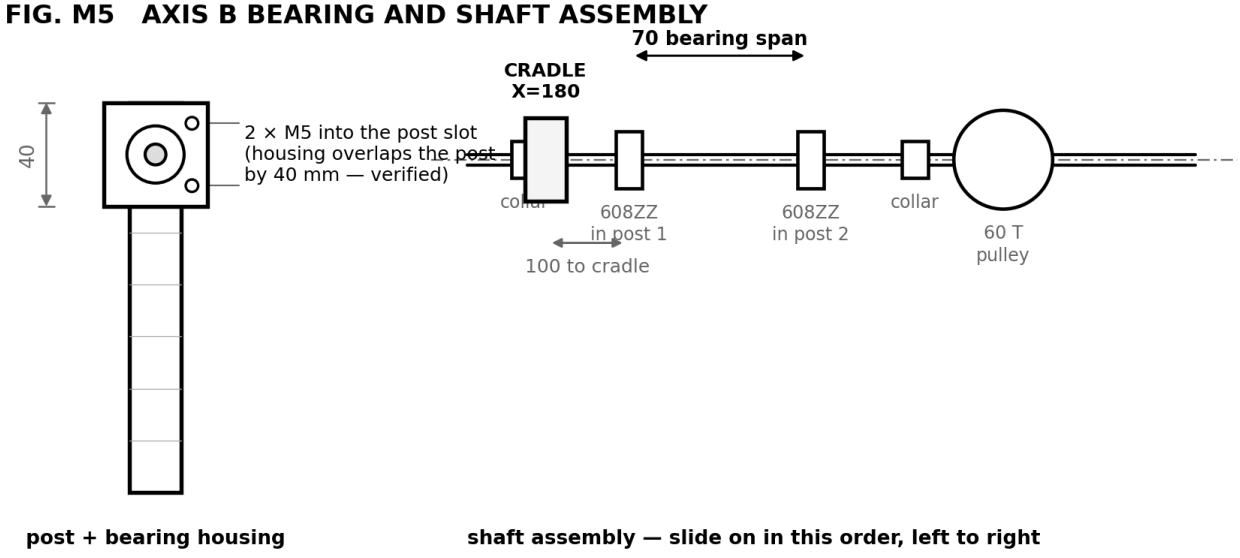


FIG. M5 — Bearing housing fixing (left) and the shaft assembly order (right). Slide the components on in the sequence shown; several cannot be fitted afterwards without dismantling.

11. Press one 608ZZ bearing into each printed housing. The fit should need firm thumb pressure but not a hammer. [B06, C11]

WARNING — If the bearing drops in loosely, the housing has printed undersize on the outer diameter. Add one wrap of PTFE tape or a smear of retaining compound. A loose outer race will creep and knock.

CHECK [ ] Bearing seated square and flush. It spins freely by finger with no gritty feel.

12. Bolt one housing to the inboard face of each post with two M5 bolts into the post slot, so the bore centre sits at Z = 200 mm. [C05, C06]

CHECK [ ] Both bore centres at Z = 200 ± 0.5 mm, measured from the bench.

13. Check coaxiality by sliding the 8 mm shaft through both bearings. [B07]

WARNING — If the shaft binds, do not force it. Slacken one housing, insert the shaft, then re-tighten that housing with the shaft in place — this self-aligns the pair.

CHECK [ ] The shaft passes through both bores under light finger pressure, with no forcing and no point at which it binds.

14. Withdraw the shaft. Now slide components on in this order, from the cradle end: shaft collar, then through bearing 1, then through bearing 2, then shaft collar, then the 60 T pulley. Position the shaft so it spans X = 150 to X = 400. [B07, B08, B02]

CHECK [ ] Shaft ends at X = 150 and X = 400 ± 2 mm. Pulley at X = 385.

15. Set both collars against the outer faces of the bearings, leaving about 0.2 mm of axial float. Threadlock and tighten the collar grub screws. [I07]

WARNING — Clamping the collars hard against the bearings pre-loads them axially and the shaft will feel stiff. Leave the small float.

CHECK [ ] The shaft turns freely but has no perceptible in-and-out movement when pushed and pulled.

16. Threadlock and tighten the 60 T pulley grub screw onto the shaft flat.

CHECK [ ] Pulley does not rotate on the shaft when twisted firmly by hand.

17. Mount Motor B to the base on its slotted L-bracket at the far end, shaft pointing inboard along −X, with its 20 T pulley coplanar with the 60 T pulley at X = 385 and its axis at Z = 90.7 mm. [A01, C10, B01]

CHECK [ ] A straight edge laid across both pulley faces touches both — the pulleys are coplanar to within 1 mm.

FIG. M8 BELT DRIVE AND TENSIONING

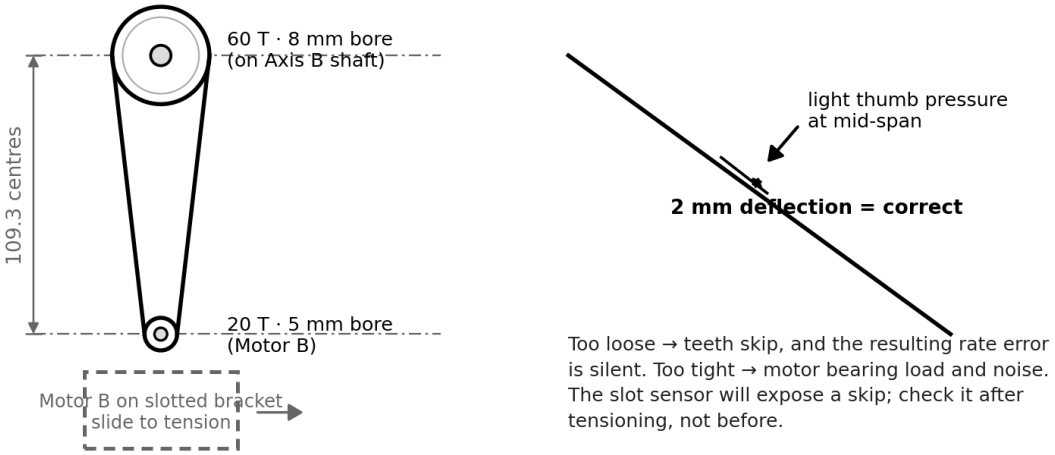


FIG. M8 — Belt geometry and the tension test. A 300 mm (150-tooth) belt gives 109.3 mm between centres at the 3:1 ratio.

18. Fit the 300 mm GT2 belt over both pulleys and slide Motor B on its slots until the belt is tensioned. [B03]

CHECK [ ] Belt deflects about 2 mm at mid-span under light thumb pressure. Tighten the bracket bolts and re-check.

19. Rotate the 20 T pulley by hand through several full sweeps of the shaft.

CHECK [ ] The shaft turns smoothly through the full ±60° range with no binding, no belt skip and no audible roughness from either bearing.

VERIFICATION GATE G2 — AXIS B

Bearings coaxial; shaft turns freely with no bind and no axial float beyond 0.2 mm. Pulleys coplanar. Belt tensioned to about 2 mm mid-span deflection. Full ±60° sweep by hand is smooth and silent.

If this gate fails, stop. A tight or notchy axis will not improve once loaded. Belt skip produces a silent rate error that would corrupt every measurement without ever announcing itself.

PHASE 3

Axis A — cradle, motor, hub and ground-truth sensor

FIG. M6 CRADLE — PRINTED PART (PETG, 4 perimeters, 40 % infill, printed arms-up; dimensions in mm)

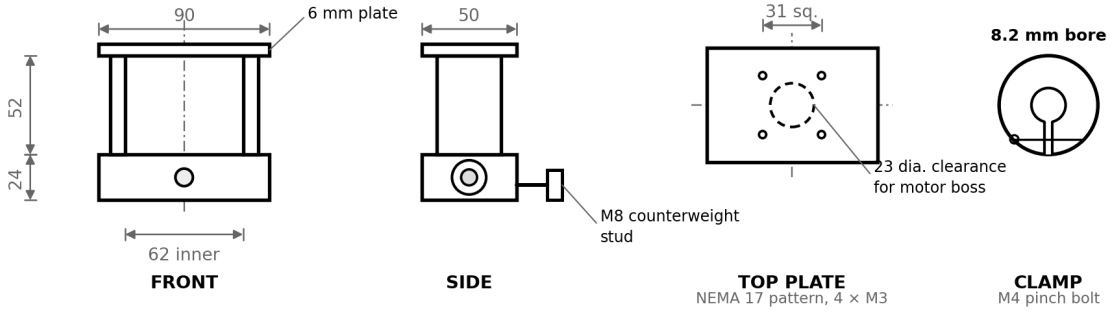


FIG. M6 — Cradle printed part. Note the 8.2 mm bore: the split clamp must be able to close onto a 8.0 mm shaft. A bore printed at exactly 8.0 mm cannot clamp and will slip.

20. Inspect the printed cradle. Clean the split-clamp slot and test that an M4 bolt runs freely through the pinch hole. [C11]

CHECK [ ] Clamp closes visibly when the M4 bolt is tightened; the slot is not fused shut by stringing.

21. Test-fit the cradle bore onto a spare length of 8 mm rod.

WARNING — If the bore is tight, ream or sand it — do not force it. Splitting the clamp on assembly is a common and irreversible failure.

CHECK [ ] Cradle slides on with the pinch bolt loose, and locks solidly when tightened.

FIG. M7 CRADLE ASSEMBLY SEQUENCE

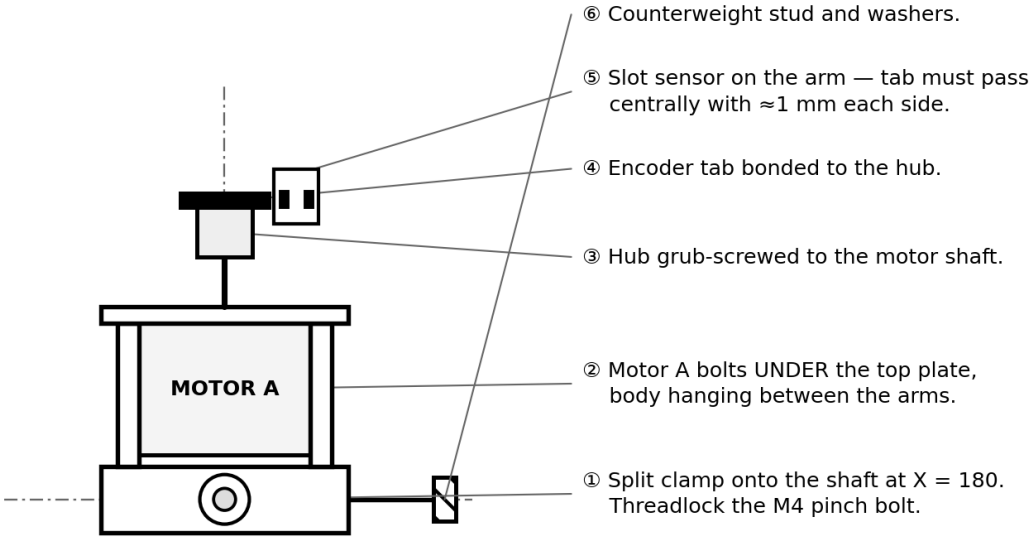


FIG. M7 — Cradle assembly sequence. Motor A bolts under the top plate with its body hanging between the arms; the hub, encoder tab and slot sensor build up from there.

22. Bolt Motor A to the underside of the cradle top plate with four M3 bolts, shaft pointing up through the clearance hole. [A01, C07, C08, C09]

CHECK [ ] Motor body hangs between the arms with about 2 mm clearance on each side. Shaft protrudes about 18 mm above the top plate.

23. Fit the aluminium hub to the motor shaft. Align its grub screw with the shaft flat, threadlock, and tighten. [B09, G04, I07]

> **CHECK** [ ] Hub does not rotate on the shaft under firm hand torque. Hub face runs true — no visible wobble when the shaft is turned.

24. Bond the encoder tab to the hub, projecting radially. [D03]

> **CHECK** [ ] Tab is rigid, square to the hub face, and does not flex when flicked.

25. Mount the slot sensor to the cradle arm so the tab passes through the slot once per revolution. [D01, C07, C08]

> **CHECK** [ ] Tab passes centrally with roughly 1 mm clearance on both sides, and does not touch the sensor body at any point in the revolution.

26. Slide the cradle onto the shaft at X = 180, orient the arms vertically, and tighten the pinch bolt with threadlock.

> **CHECK** [ ] Cradle at X = 180 ± 2 mm, arms vertical, and it does not slip when the target end is loaded by hand.

27. Fit the M8 counterweight stud to the cradle on the opposite side of the shaft from the motor, with a few washers to begin. [C12, C13]

> **CHECK** [ ] Stud secure and clear of the sweep path.

FIG. M9 CRADLE BALANCE — PERFORM WITH POWER OFF AND BELT RELEASED

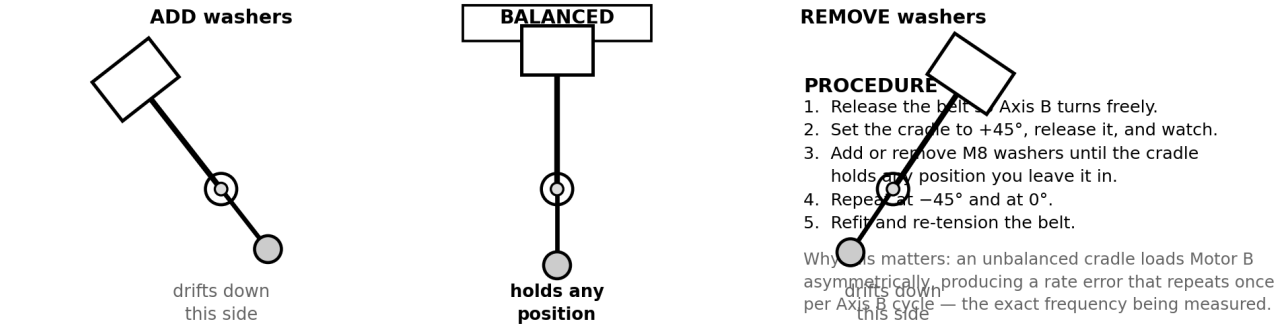


FIG. M9 — Balance procedure. An unbalanced cradle produces a rate error at exactly the Axis B frequency, which is one of the frequencies being measured.

28. Release the belt from the 20 T pulley so Axis B turns freely. Fit a target model to the hub.

> **CHECK** [ ] Axis B rotates freely by hand with the belt off.

29. Balance the cradle: set it to +45°, release, and observe. Add or remove M8 washers until it holds any position you leave it in. Repeat at −45° and at 0°.

> **WARNING —** Do not skip this. An unbalanced cradle loads Motor B asymmetrically through the sweep, producing a periodic rate error at the Axis B frequency — indistinguishable, in the light curve, from real signal.

> **CHECK** [ ] The cradle holds all three positions with no drift over ten seconds.

30. Refit and re-tension the belt.

> **CHECK** [ ] Tension check repeats: about 2 mm deflection at mid-span.

31. Route the Motor A and sensor cables in spiral wrap, forming a service loop near the shaft, clipped so nothing can snag at either sweep extreme. [H08, H10]

> **CHECK** [ ] Sweep the cradle by hand through the full ±60° while watching the cable. It flexes freely, never goes taut, and never touches a moving part.

VERIFICATION GATE G3 — CRADLE

Cradle balanced at −45°, 0° and +45°. Hub and tab rigid and true. Tab passes centrally through the sensor once per revolution. Cable loop flexes freely at both sweep extremes.

If this gate fails, stop. Imbalance and cable snag both produce faults that look like data rather than like errors. They are far cheaper to fix now than to diagnose in a light curve.

PHASE 4

Electrical — harness, drivers and first power

Build the power harness completely, and prove it dead, before a single driver or motor is connected. This ordering is what prevents the two most common destructive failures in this class of build.

FIG. M10 12 V POWER HARNESS — FUSE AND ISOLATION SWITCH

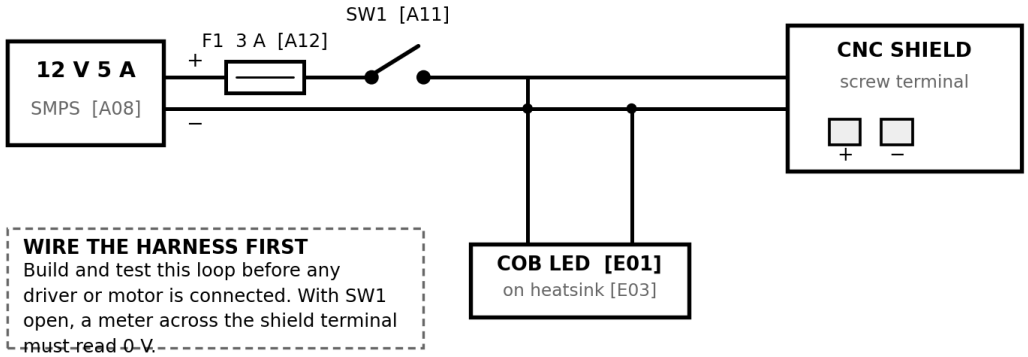


FIG. M10 — The 12 V harness. F1 and SW1 go in the positive line between the supply and the shield terminal. The COB LED taps the same rail downstream of both.

32. Wire the 12 V harness: supply positive → F1 fuse holder → SW1 switch → shield screw terminal positive. Supply negative runs straight to the shield terminal negative. [A08, A09, A11, A12, H01, H06, H07]

CHECK [ ] Continuity confirmed with the meter along each leg, supply unplugged.

33. Fit a 3 A fuse. Plug in the supply with SW1 open, and measure across the shield terminals.

WARNING — If the meter reads 12 V with SW1 open, the switch is wired across the wrong leg. Fix this before proceeding — an isolation switch that does not isolate is worse than none, because you will trust it.

CHECK [ ] Meter reads 0 V with SW1 open, and 12 V ± 0.5 V with SW1 closed.

34. Open SW1 and unplug the supply. Mount the Arduino and CNC shield to the base on nylon standoffs, clear of the aluminium frame. [A04, A06, A13]

CHECK [ ] No part of the board underside touches metal. Boards do not flex when the USB cable is inserted.

(a) COIL-PAIR IDENTIFICATION — before any connector is made **(b) DRIVER CURRENT SETTING**

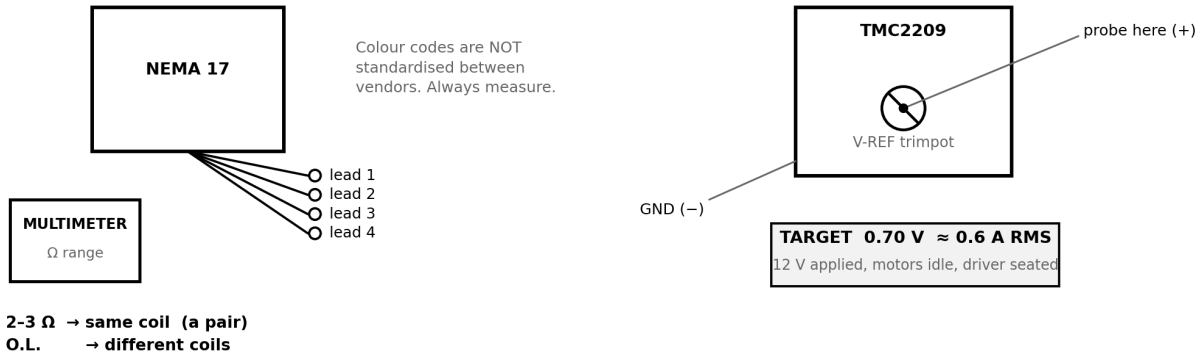


FIG. M11 — (a) Coil-pair identification with a multimeter. (b) V-REF measurement point on the driver.

35. With everything unpowered, identify each motor's coil pairs: probe pairs of leads on the ohms range. Roughly 2–3 Ω means one coil; open circuit means different coils. Label the leads.

WARNING — Motor lead colour codes are not standardised between vendors. Never assume a colour convention — always measure. A mis-paired motor buzzes, heats and does not turn.

CHECK [ ] Two pairs identified per motor, each reading 2–3 Ω, with open circuit between pairs.

36. Crimp JST connectors onto both motor harnesses, keeping each identified pair together on adjacent pins. [H05]

CHECK [ ] Continuity from each connector pin to its correct motor lead. No shorts between pins.

37. Fit both TMC2209 drivers into the X and Y sockets. Check orientation against the shield silkscreen — the enable pin marking on the driver must match the socket. [A02]

WARNING — A StepStick inserted backwards is destroyed the instant power is applied, and can take the shield with it. Check twice. This is the single most expensive mistake available in this build.

CHECK [ ] Both drivers seated fully, all pins in their sockets, orientation matching the silkscreen.

38. Install the microstepping jumpers for 1/16 on both sockets — MS1 and MS2 both high. Confirm the jumper-to-pin mapping against your specific driver board's markings.

CHECK [ ] Two jumpers fitted per driver socket, in the positions your board's documentation specifies for 1/16.

39. Fit the adhesive heatsinks to both drivers. [A07]

CHECK [ ] Heatsinks bonded flat to the driver ICs, not bridging any adjacent components.

40. Connect both motors. Close SW1 and set each driver's V-REF trimpot to 0.70 V, measured between the trimpot wiper and ground with motors idle. [I07]

WARNING — Open SW1 before connecting or disconnecting any motor. Back-EMF from a coil disconnecting under power destroys the driver.

CHECK [ ] Both drivers read 0.70 ± 0.03 V.

41. Leave the system powered and idle for two minutes, then touch each driver heatsink briefly.

CHECK [ ] Heatsinks are warm but can be touched comfortably. If either is too hot to hold, open SW1 and reduce V-REF toward 0.55 V.

VERIFICATION GATE G4 — POWER

SW1 proves the rail dead when open. Fuse fitted. Coil pairs identified by measurement. Both drivers correctly oriented, jumpered for 1/16, heatsunk, and set to 0.70 V. Drivers warm, not hot, after two minutes idle.

If this gate fails, stop. Every failure this gate catches is destructive and irreversible. Nothing downstream is worth risking a driver or a shield to save ten minutes here.

PHASE 5

Firmware and commanded motion

The controller sketch must do four things and nothing else. Anything additional in the main loop disturbs step timing and degrades the very measurement the instrument exists to make.

Function	Requirement
Accept rates	Two values over serial at 115200 baud, in microsteps per second, one per axis.
Drive Axis A	Continuous rotation at the commanded rate. Constant velocity — no acceleration profile once running.
Drive Axis B	Triangular sweep between ±3200 microsteps from centre, reversing at each limit.
Report truth	Poll the sensor pin every loop iteration; on a rising edge, print the `micros()` value.

42. Upload the sketch with both motors disconnected and SW1 open.

CHECK [ ] Sketch compiles and uploads without error. Serial monitor opens at 115200 baud and shows the startup banner.

43. Connect Motor A only. Close SW1 and command a slow rate — about 400 microsteps per second.

CHECK [ ] Motor A turns smoothly and continuously in one direction, with no stalling, buzzing or missed steps.

44. Increase Motor A to the working rate of 1730 microsteps per second, which is one revolution every 1.85 s.

CHECK [ ] Time twenty revolutions with a stopwatch: 37.0 ± 0.5 s. Motion is smooth and near-silent.

45. Open SW1. Connect Motor B. Close SW1 and command the Axis B sweep at 561 microsteps per second.

WARNING — Watch the first full sweep with a hand near SW1. If the cradle runs past ±60° and strikes anything, open SW1 immediately and check the sweep-limit constant in the sketch.

CHECK [ ] The cradle sweeps between the limits and reverses cleanly. One full cycle takes 11.4 ± 0.3 s.

46. Run both axes together for five minutes.

CHECK [ ] Both continue at their commanded rates. No stall, no belt skip, no cable snag, no rising motor temperature beyond warm.

47. Watch the serial stream from the slot sensor while Axis A runs.

CHECK [ ] One timestamp per revolution, with intervals constant to better than 1 % — for a 1.85 s period, successive differences of 1,850,000 ± 15,000 µs.

48. Compute the period from twenty consecutive sensor intervals and compare against the commanded rate.

CHECK [ ] Measured period within 0.5 % of 1.85 s. Any larger discrepancy means belt skip, a slipping grub screw, or a wrong microstep jumper.

VERIFICATION GATE G5 — MOTION

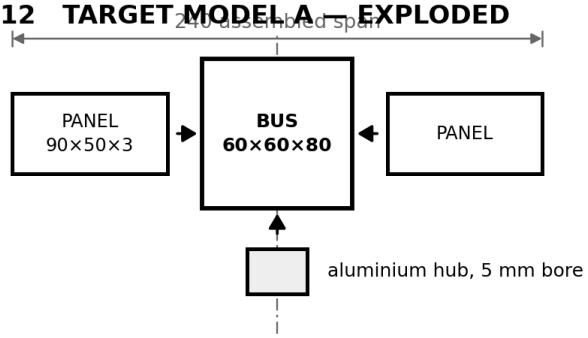
Both axes run at commanded rates, verified by stopwatch. Five-minute continuous run completes without stall, skip or snag. Sensor streams one clean pulse per revolution, intervals constant to better than 1 %, and the derived period matches the command to within 0.5 %.

If this gate fails, stop. This gate is where a silent mechanical fault — belt skip, slipping hub, wrong microstepping — finally becomes visible. Every one of those faults would otherwise appear later disguised as a measurement result.

PHASE 6

Target models

FIG. M12 TARGET MODEL A — EXPLODED



SURFACING — THIS IS NOT OPTIONAL

Bare PLA is a uniform diffuser in every direction. An untreated target produces a low-contrast, featureless curve with almost nothing to invert.

BUS — wrap in aluminised mylar tape, deliberately creased. Creases give the irregular multi-lobed scatter of real multilayer insulation.

PANEL — face with blue acrylic or blue cellophane over the printed substrate. This is the specular surface that produces the glints.

HUB — metal, grub-screwed. Printed bores strip and the resulting eccentricity is hard to diagnose.

FIG. M12 — Target Model A exploded, with the surfacing requirements. The surfacing is not cosmetic: it creates the specular-to-diffuse contrast that the entire measurement depends on.

49. Clean up the printed bus and panels — remove brim, elephant's foot and any stringing from the mating faces. [I05]

CHECK [ ] Panels sit flush against the bus faces with no rocking.

50. Wrap the bus in aluminised mylar tape, deliberately creasing it as you go. [G02]

WARNING — A perfectly smooth wrap behaves as a single mirror facet and produces one narrow unrealistic spike. Creases are what make it resemble real multilayer insulation.

CHECK [ ] Whole bus covered. Surface shows irregular multi-lobed reflections rather than one flat mirror face — turn it under a lamp and confirm the highlight breaks up.

51. Face each panel with blue acrylic offcut or blue cellophane over the printed substrate, bonded flat with foam tape. [G03, G05]

CHECK [ ] Panel faces are flat and specular. Held at a glancing angle under a lamp, each throws a clean directional highlight.

52. Attach both panels to opposite faces of the bus.

CHECK [ ] Assembled span 240 ± 3 mm. Panels coplanar — sight along the span and confirm neither is cocked.

53. Fit the metal hub into the base of the bus and secure it. [B09]

CHECK [ ] Hub is square to the bus and does not rotate within it.

54. Balance the assembled target on the hub: rest the hub on a horizontal pencil or rod and let it settle.

CHECK [ ] The target shows no strong preference for one orientation. Correct any imbalance by shifting tape mass, not by adding weight to a panel.

55. Repeat steps 49 to 54 for Model B — single panel plus the offset dish.

CHECK [ ] Model B assembled, span 238 ± 3 mm, and visibly asymmetric about the spin axis.

VERIFICATION GATE G6 — TARGETS

Both models assembled, surfaced, and balanced on the hub. Bus scatter is multi-lobed; panel faces are cleanly specular. Spans within tolerance.

If this gate fails, stop. An untreated or unbalanced target produces a low-contrast, vibration-blurred light curve. No amount of software effort recovers information that was never in the signal.

PHASE 7

Enclosure, illumination and camera

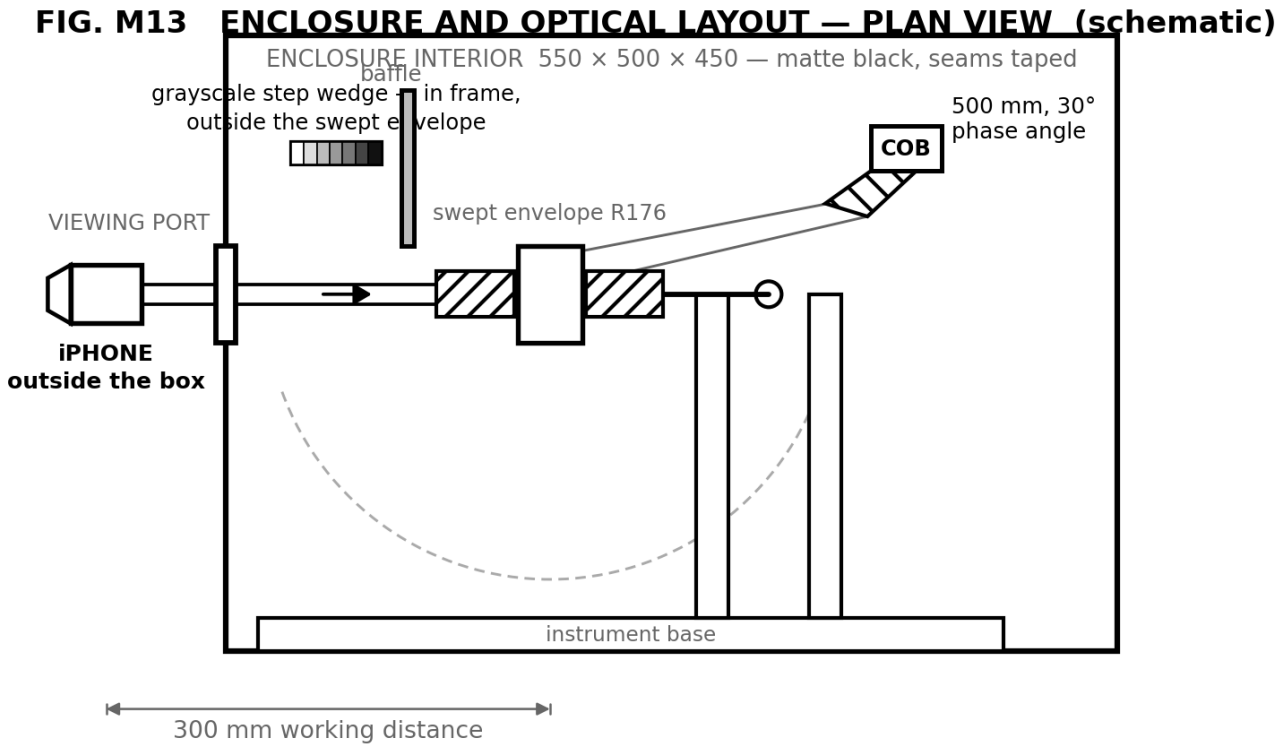


FIG. M13 — Enclosure and optical layout. The camera sits outside the box and views through a port; the baffle blocks every direct path from lamp to lens.

56. Build or trim the enclosure to an interior of at least 550 × 500 × 450 mm and line the interior with black chart paper. Paint any reflective hardware inside the field of view matte black. [E06, E07, E10]

CHECK [ ] No bare cardboard, no bright metal and no printed logo visible from the camera position.

57. Cut a viewing port in the −X face, centred on Z = 200 mm, about 60 × 60 mm.

CHECK [ ] Looking through the port from 300 mm, the whole swept envelope of the target is visible with margin on all sides.

58. Bond the COB LED to its heatsink with thermal adhesive and wire it to the 12 V rail downstream of F1 and SW1. [E01, E03, E04]

WARNING — Never power the COB without its heatsink attached, even briefly. An unheatsunk 10 W COB reaches damaging temperature within about a minute, and a partially degraded LED then droops in output during every subsequent recording.

CHECK [ ] LED lights when SW1 is closed. Heatsink is warm, not hot, after five minutes.

59. Fit the snoot to the lamp and position it at about 500 mm from the target, at 30° from the viewing axis. [E06, E08]

CHECK [ ] The illuminated patch covers the swept envelope and does not spill onto the enclosure walls.

60. Fit the stray-light baffle between the lamp and the port. [E08]

CHECK [ ] From the port, with the target removed, no part of the LED emitter is directly visible.

61. Fix the grayscale step wedge inside the frame, outside the swept envelope. [E09]

CHECK [ ] Wedge visible in the camera frame at all target attitudes, and never shadowed by the target.

62. Set the tripod at 300 mm working distance, level the phone clamp, and mark the tripod foot positions on the floor with tape. [F01, F02, F03]

CHECK [ ] Camera framing repeats after the tripod is moved away and returned to the marks.

63. Seal all enclosure seams with tape, and work with room lights off from here on. [I04]

CHECK [ ] With the lamp off and the room dark, a 5 s recording shows a near-black frame.

Flicker verification

64. With the lamp on and the target removed, record 5 s at 60 fps. Extract frames and plot the frame-mean brightness against frame number.

WARNING — A flickering lamp injects a periodic signal that is indistinguishable from a real detection. It is the single most dangerous fault in this instrument because it produces a confident wrong answer rather than an obvious failure.

CHECK [ ] The trace is flat to within about 1 %. Any periodic ripple means the lamp is not running on clean DC — find and fix it before recording any data.

Stray-light verification

65. With the lamp on and the target still removed, record 10 s and compute the mean frame level. Then refit the target, run Axis A, and record the peak level of a glint.

CHECK [ ] Stray-light mean is below 2 % of the glint peak. If not, find the leak — tape seams, add baffling, and re-test.

VERIFICATION GATE G7 — OPTICS

Enclosure light-tight. Lamp flat to within 1 % over 5 s with no periodic ripple. Stray-light floor below 2 % of glint peak. Step wedge in frame and never shadowed. Camera geometry repeatable from tripod marks.

If this gate fails, stop. Both failures this gate catches — flicker and stray light — corrupt the data silently. One invents a signal; the other suppresses the real one.

PHASE 8

Camera setup and first light

66. Install Blackmagic Camera. In iOS Settings → Camera, turn Smart HDR off.

WARNING — Smart HDR applies per-frame non-linear tone mapping. Left on, it makes photometric measurement meaningless while producing footage that looks perfectly normal.

CHECK [ ] Smart HDR shows as off in Settings.

67. Set 1080p at 60 fps. Frame the target with margin for the full sweep. Lock focus at the nominal attitude.

CHECK [ ] Target and step wedge both in frame at −60°, 0° and +60°. Focus does not hunt when the target moves.

68. Run Axis A. Adjust ISO and shutter until the brightest glint peaks at about 85 % of full scale, then lock ISO, shutter and white balance.

WARNING — A clipped glint cannot be recovered by any post-processing. The specular amplitude carries most of the geometric information, so a clipped recording is worthless for shape inversion even though it still yields a period.

CHECK [ ] Histogram shows the glint peak near 85 % with no clipping. All three locks engaged.

69. Allow the lamp to reach thermal equilibrium for ten minutes before recording data.

CHECK [ ] Two 10 s recordings taken five minutes apart have frame-mean levels within 1 % of each other.

70. Record run V1: Model A, Axis A only at 1.85 s, Axis B held stationary, 60 s. Start the serial log at the same moment.

WARNING — If the frame count does not equal duration × 60, the recording is variable frame rate and the time base is invalid. Re-record — this cannot be corrected afterwards.

CHECK [ ] Video and serial log both cover the full 60 s. Extracted frame count equals 3600 exactly.

71. Plot the light curve from V1.

CHECK [ ] Clear periodic modulation visible by eye, with two glints per revolution for Model A.

72. Compute the period from V1 by Lomb–Scargle and compare against the slot-sensor period from the serial log.

CHECK [ ] Optical and sensor periods agree to better than 1 %, after resolving the factor-of-two harmonic ambiguity.

73. Record runs V2 to V5 per SDC-001 Section 13.3, each 180 s.

CHECK [ ] All four runs complete with correct frame counts and matching serial logs.

VERIFICATION GATE G8 — FIRST DATA

V1 recorded at constant frame rate with a matching serial log. Light curve shows clear modulation. Optically derived period agrees with the independent slot-sensor period to better than 1 %. Runs V2–V5 captured.

If this gate fails, stop. This is the gate that proves the instrument works. Agreement between two physically independent measurements of the same quantity is the whole claim — everything before this point exists to make it possible.

HOW IT WORKS

A short account of what each part of the machine is for, to be read once the build is complete — or by anyone asking why it is shaped the way it is.

The measurement

A dead satellite emits no light. It reflects sunlight, and because it is not spherical, the fraction it sends toward an observer changes as it tumbles. A flat solar panel turning face-on produces a sharp glint; the same panel edge-on returns almost nothing. Recording brightness against time gives a periodic trace whose rhythm encodes the rotation rate, and whose shape encodes something about the geometry.

This instrument reproduces that situation at bench scale, with one crucial addition: **the answer is known in advance**. The tumble is commanded in firmware and measured by a sensor that never touches the optical path. The analysis can therefore be graded rather than merely admired.

Why two axes

A target spinning about a single axis produces a near-sinusoidal curve that any Fourier transform solves immediately. Real derelicts tumble about a non-principal axis, producing two incommensurate frequencies that beat against one another. Axis A spins the target; Axis B rocks the whole cradle, sweeping the spin axis through the observer's frame. The result is a genuinely difficult inversion problem rather than a trivial one.

The rates are chosen so that $P_B/P_A = 6.16$ — deliberately not an integer, so the composite pattern never repeats within a recording. If the ratio were an exact integer the two frequencies would lock, the curve would become simply periodic, and the inversion would appear to succeed while actually solving a much easier problem.

Why a rocking cradle rather than a full gimbal

Continuous rotation of the second axis would require power and signal to cross a rotating joint, which means a slip ring — friction, contact noise on the sensor line, and another component to fail. Limiting Axis B to ±60° lets the cables pass through a service loop that flexes rather than wraps. The kinematic benefit is kept; a failure mode is removed.

Why the bearings sit on one side

This is the change from SDC-001. With the shaft supported on both sides of the cradle, the target's swept envelope — a circle of radius 176 mm — overlaps the posts. The cantilever arrangement puts both bearings behind the sweep plane, leaving the target a clear 350 mm of clear air. The penalty is a 100 mm overhang, which the calculation in Section 1.3 shows produces 8.7 MPa of shaft stress against a 250 MPa yield, and 0.06 mm of deflection. Both are negligible.

Why the illumination is DC

Mains-driven LEDs modulate at 100 Hz. Sampling at 60 fps produces a beat between the two, which appears in the light curve as a strong, stable, entirely spurious periodicity. The danger is not that it fails — it is that it succeeds, producing a confident measurement of nothing. A COB module run from the same regulated DC rail as the motors removes the problem for ₹250.

Why TMC2209 rather than the cheaper driver

An A4988 at coarse microstepping cogs — the instantaneous rotation rate varies periodically at the step frequency. That is, again, a periodic signal contaminating a periodic measurement. The TMC2209 in StealthChop with 1/16 microstepping and internal 256-step interpolation keeps this artefact well below the measurement floor.

Why an eighty-rupee sensor is treated as critical

The slot sensor measures the true rotation period through a channel with no camera, no codec and no photometry in it. Without it, the project can demonstrate that its software produces an answer. With it, the project can demonstrate that the answer is *correct*. That is the entire difference between a demonstration and

a measurement, and it costs less than the cable ties.

FIG. M14 THE EIGHT VERIFICATION GATES

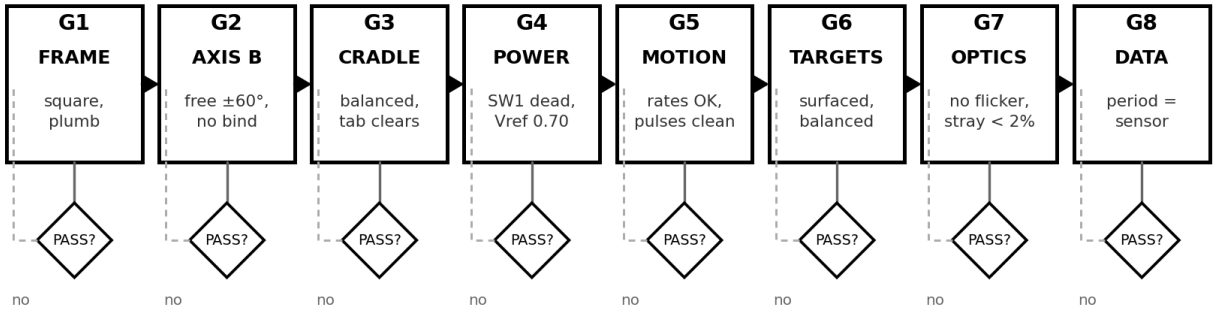


FIG. M14 — The seven verification gates. Each exists because the fault it catches is substantially more expensive to find later than to find here.

APPENDIX A

Verification log

Tick and date each gate as it passes. Record the measured value where one is called for — the numbers are useful evidence in the final write-up, and a gate recorded as 'passed' with no number behind it is worth very little.

Gate	Criterion	Measured	Pass	Date
Pre-build	All BOM items present; six lengths cut and square			
G1 Frame	Diagonal difference (target < 1 mm)			
	Post tops at Z = 220 mm			
	Post centre spacing top and bottom (70 ± 0.5 mm)			
G2 Axis B	Bore centres at Z = 200 ± 0.5 mm			
	Shaft passes both bearings without force			
	Axial float (target ≈ 0.2 mm)			
	Belt mid-span deflection (target ≈ 2 mm)			
G3 Cradle	Holds position at −45°, 0°, +45°			
	Tab clearance in sensor slot (≈1 mm each side)			
	Cable loop free at both sweep extremes			
G4 Power	Rail voltage with SW1 open (must be 0 V)			
	Rail voltage with SW1 closed (12 ± 0.5 V)			
	Motor A coil resistance (2–3 Ω per pair)			
	Motor B coil resistance (2–3 Ω per pair)			
	Driver X V-REF (0.70 ± 0.03 V)			
	Driver Y V-REF (0.70 ± 0.03 V)			
G5 Motion	Twenty Axis A revolutions (37.0 ± 0.5 s)			
	Axis B full cycle (11.4 ± 0.3 s)			
	Sensor interval spread over 20 pulses (< 1 %)			
	Five-minute continuous run completed			
G6 Targets	Model A span (240 ± 3 mm)			
	Model B span (238 ± 3 mm)			
	Both balanced on the hub			
G7 Optics	Lamp flatness over 5 s (< 1 % ripple)			
	Stray-light mean as % of glint peak (< 2 %)			
	Framing repeats from tripod marks			
G8 First data	V1 frame count (must be exactly 3600)			
	Optical period from V1			
	Slot-sensor period from V1			
	Agreement (target < 1 %)			

APPENDIX B

Key dimensions — quick reference

FIG. M3 GENERAL ARRANGEMENT — ELEVATION (viewed along the shaft axis is Fig. M1b)

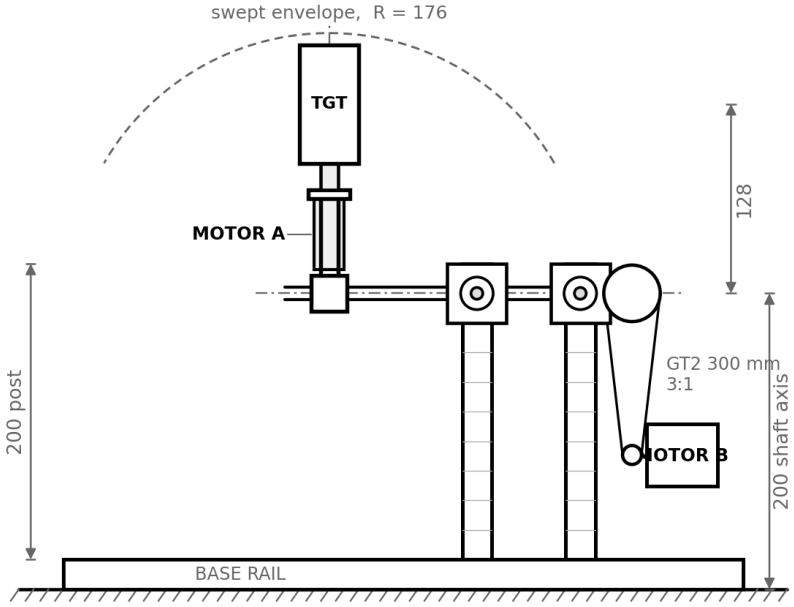


FIG. M3 — General arrangement in elevation, viewed across the shaft. The swept envelope is shown dashed.

Feature	Value	Set at step
Base frame, outer	460 × 200 mm	6
Post height	200 mm (tops at Z = 220)	9
Post positions	X = 280 and X = 350	9
Bearing bore centres	Z = 200 mm	12
Bearing span	70 mm	12
Shaft	8 mm dia., X = 150 to 400	14
Cradle position	X = 180 (100 mm overhang)	26
Shaft axis to target centre	128 mm	by cradle geometry
Target span, Model A	240 mm	52
Swept envelope radius	176 mm	consequence
Lowest swept point	Z = 160 mm (140 mm above rail top)	consequence
60 T pulley	X = 385	14
Motor B axis	Z = 90.7 mm	17
Belt centres	109.3 mm, 300 mm belt, 3:1	18
Axis B sweep	±60° = ±3200 microsteps	firmware
Axis A period	1.85 s = 1730 microsteps/s	firmware
Axis B period	11.4 s = 561 microsteps/s	firmware
Camera working distance	300 mm	62
Phase angle	30°	59
Enclosure interior	550 × 500 × 450 mm minimum	56

IF YOU CHANGE ONE DIMENSION, RE-CHECK THE ENVELOPE

The swept envelope radius is the sum of the shaft-to-target distance and the target half-span. Increasing either — a longer cradle, a wider target — increases the radius and can bring the target back into contact with the posts or the bench. The clearances quoted here hold for 128 mm and 120 mm respectively. Any change to either must be re-checked against the post position at X = 280 and the rail top at Z = 20.

APPENDIX C

Build troubleshooting

Symptom	Cause	Fix
Shaft binds in the bearings	Housings not coaxial	Slacken one housing, insert the shaft, re-tighten with the shaft in place. This self-aligns the pair.
Bearing drops loosely into its housing	Housing printed undersize on the outer diameter	One wrap of PTFE tape on the outer race, or a smear of retaining compound.
Cradle clamp will not grip the shaft	Bore printed at 8.0 mm instead of 8.2 mm, so the split cannot close	Ream or sand the bore until the clamp visibly closes when tightened.
Cradle slips under load	Pinch bolt under-tightened, or bore polished	Threadlock and re-tighten. If it still slips, add a shallow flat to the shaft.
Cradle drifts to one side when released	Imbalance	Add or remove M8 washers on the counterweight stud until it holds all three test positions.
Motor buzzes but does not turn	Coil pairs mis-wired	Open SW1. Re-identify pairs with the meter and rewire.
Motor turns the wrong way	Coil polarity	Reverse one pair only, or invert DIR in firmware.
Driver very hot within a minute	V-REF too high	Open SW1, reduce toward 0.55 V, confirm the heatsink is bonded flat.
Rail reads 12 V with SW1 open	Switch wired into the wrong leg	Rewire so SW1 interrupts the positive line between F1 and the shield terminal.
Fuse blows on first power-up	Short in the harness, or a driver inserted backwards	Open SW1. Check driver orientation against the silkscreen, then ring out the harness before refusing.
Belt skips under load	Tension too low, or pulleys not coplanar	Re-tension to 2 mm mid-span; lay a straight edge across both pulley faces to check coplanarity.
Sensor gives no pulses	Tab misaligned, or module powered from 12 V	Confirm the tab passes through the slot and that the module is on the 5 V rail.
Sensor intervals erratic	Loose tab, or electrical noise on the signal line	Threadlock the tab fixing; shorten the sensor lead and route it away from motor cables.
Frame-mean brightness ripples with no target present	Lamp flicker	The lamp is not on clean DC. Confirm it is the DC COB on the regulated rail and not a mains fixture.
Light curve modulation much shallower than expected	Stray light, or an unsurfaced target	Run the 2 % floor test. If it passes, the target needs its specular facing refreshed.
Extracted frame count ≠ duration × 60	Variable frame rate	Re-record with a fixed frame rate. This cannot be corrected in post-processing.
Glint peaks flat-topped	Sensor clipping	Reduce ISO or shorten shutter, then re-record. Clipped data is unrecoverable.

END OF MANUAL — SDC-002 REV A